

ECON 712: Applied Econometrics

Lecture 1: Introduction

Empirical Questions

- The study of Applied Econometrics is based on establishing causal relationships using data
 - ▶ Therefore, an applied econometric project must have well-defined empirical question based on a causal relationship
- The best way to approach causal relationships is to think about counterfactuals/potential outcomes, which can be modelled as follows in the simplest case where
 - ▶ there are two options, no treatment (0) or treatment (1)
 - ▶ there is a unique constant effect for all individuals i

$$y_{1i} - y_{0i} = \rho$$

Empirical Questions

- The potential outcome framework is a very helpful way of defining causal relationships because
 1. it forces us to be very specific about the relationship we are trying to uncover
 2. it helps in identifying which variables are potentially helpful to establish the relationship
 - ▶ Also, it helps with identifying which questions are FUQs
- The potential outcome framework can also be extended, with additional structure, to marginal effects as we usually study in regression analysis

Identifying the effect in the data

- The issue that a researcher faces is that s/he only observes:

$$y_i = y_{0i} + (y_{1i} - y_{0i})D_i \quad , \quad D_i = \begin{cases} 0 & i \text{ did not take the treatment} \\ 1 & i \text{ took the treatment} \end{cases}$$

- So, the question is how could the causal relationship be identified from the data?

- ▶ Empirically, the outcome variable, y_i is a random variable, so a natural comparison is between observed conditional means:

$$E[y_i|D_i = 1] - E[y_i|D_i = 0] = \underbrace{(E[y_{1i} - y_{0i}|D_i = 1])}_{\rho^{ATT}} + \underbrace{(E[y_{0i}|D_i = 1] - E[y_{0i}|D_i = 0])}_{\text{selection Bias}}$$

Example: NHIS

- We want to study whether using hospitals for primary care among an elderly and poor population yields better health outcomes
 - ▶ The idea is that hospitals are better equipped to deal with health issues in that population
 - ▶ To answer this question, we look at observational data from the National Health Interview Survey:

Group	Sample Size	Mean Health Status	Std. Error
Hospital	7,774	3.21	0.014
No hospital	90,049	3.93	0.003

Randomization

- The issue with using the survey is that individuals who are hospitalized are, on average, in much poor health (selection Bias)
- We can also use regression analysis to study this issue:

$$y_i = \underbrace{\alpha}_{E[y_{0i}]} + \underbrace{\rho}_{y_{1i}-y_{0i}} D_i + \underbrace{\epsilon_i}_{y_{0i}-E[y_{0i}]}$$

where

$$E[y_i | D_i = 1] = \alpha + \rho + E[y_{0i} - E[y_{0i}] | D_i = 1]$$

$$E[y_i | D_i = 0] = \alpha + E[y_{0i} - E[y_{0i}] | D_i = 0]$$

if randomly assigned:

$$E[y_{0i} - E[y_{0i}] | D_i = 1] = E[y_{0i} - E[y_{0i}] | D_i = 0] = 0$$

Example: Project STAR

TABLE 2.2.1
Comparison of treatment and control characteristics in the Tennessee STAR experiment

Variable	Class Size			<i>P</i> -value for equality across groups
	Small	Regular	Regular/Aide	
Free lunch	.47	.48	.50	.09
White/Asian	.68	.67	.66	.26
Age in 1985	5.44	5.43	5.42	.32
Attrition rate	.49	.52	.53	.02
Class size in kindergarten	15.10	22.40	22.80	.00
Percentile score in kindergarten	54.70	48.90	50.00	.00

Notes: Adapted from Krueger (1999), table I. The table shows means of variables by treatment status for the sample of students who entered STAR in kindergarten. The *P*-value in the last column is for the *F*-test of equality of variable means across all three groups. The free lunch variable is the fraction receiving a free lunch. The percentile score is the average percentile score on three Stanford Achievement Tests. The attrition rate is the proportion lost to follow-up before completing third grade.

Example: Project STAR

TABLE 2.2.2
Experimental estimates of the effect of class size on test scores

Explanatory Variable	(1)	(2)	(3)	(4)
Small class	4.82 (2.19)	5.37 (1.26)	5.36 (1.21)	5.37 (1.19)
Regular/aide class	.12 (2.23)	.29 (1.13)	.53 (1.09)	.31 (1.07)
White/Asian	—	—	8.35 (1.35)	8.44 (1.36)
Girl	—	—	4.48 (.63)	4.39 (.63)
Free lunch	—	—	-13.15 (.77)	-13.07 (.77)
White teacher	—	—	—	-.57 (2.10)
Teacher experience	—	—	—	.26 (.10)
Teacher Master's degree	—	—	—	-0.51 (1.06)
School fixed effects	No	Yes	Yes	Yes
R^2	.01	.25	.31	.31

Notes: Adapted from Krueger (1999), table V. The dependent variable is the Stanford Achievement Test percentile score. Robust standard errors allowing for correlated residuals within classes are shown in parentheses. The sample size is 5,681.